

Birla Institute of Technology and Science, Pilani
Instructor Solution Manual
AIMLCZG511 – Deep Neural Networks
End Semester Test (Makeup)

Semester I 2025–26

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1 Question 1: Feedforward Neural Network (16 Marks)

Input vector:

$$X = [0.1, 0.3, 0.5]$$

Weight matrices:

$$W_{ij} = \begin{bmatrix} 0.2 & 0.4 & 0.5 \\ 0.3 & 0.7 & 0.1 \\ 0.2 & 0.6 & 0.9 \end{bmatrix}$$

$$W_{jk} = \begin{bmatrix} 0.4 & 0.6 & 0.1 \\ 0.2 & 0.7 & 0.3 \\ 0.8 & 0.1 & 0.5 \end{bmatrix}$$

$$W_{kl} = \begin{bmatrix} 0.2 & 0.5 & 0.3 \\ 0.4 & 0.6 & 0.8 \\ 0.1 & 0.9 & 0.7 \end{bmatrix}$$

Bias terms are zero.

1.1 1(i) Hidden Layer 1 (ReLU) (4 Marks)

$$z^{(1)} = XW_{ij}$$

Neuron 1:

$$0.1(0.2) + 0.3(0.3) + 0.5(0.2)$$

$$= 0.02 + 0.09 + 0.10 = 0.21$$

Neuron 2:

$$0.1(0.4) + 0.3(0.7) + 0.5(0.6)$$

$$= 0.04 + 0.21 + 0.30 = 0.55$$

Neuron 3:

$$0.1(0.5) + 0.3(0.1) + 0.5(0.9)$$

$$= 0.05 + 0.03 + 0.45 = 0.53$$

ReLU activation:

$$h^{(1)} = [0.21, 0.55, 0.53]$$

Marking Scheme

Matrix multiplication setup	1
Neuron z_1 correct	1
Neuron z_2 correct	1
Neuron z_3 + ReLU	1

1.2 1(ii) Activation Functions (5 Marks)

ReLU

$$ReLU(x) = \max(0, x)$$

Advantages:

- avoids vanishing gradients
- computational efficiency
- sparse activations

Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Used for bounded representation.

Softmax

$$p_i = \frac{e^{z_i}}{\sum e^{z_j}}$$

Produces probability distribution for multi-class classification.

Marking Scheme

ReLU explanation	2
Sigmoid explanation	1
Softmax explanation	2

1.3 1(iii) Cross Entropy Loss (3 Marks)

Loss function:

$$L = - \sum y_i \log(p_i)$$

Advantages over MSE:

- stronger gradients
- probabilistic interpretation
- faster convergence

Limitations of MSE:

- slower convergence
- weak gradient signal

1.4 1(iv) Gradient Flow Analysis (4 Marks)

Largest gradient:

Output layer (closest to loss).

Smallest gradient:

Hidden Layer 1 due to:

- distance from loss
- sigmoid derivative < 1
- gradient shrinkage

2 Question 2(A): RNN Architectures (12 Marks)

2.1 Vanishing Gradient Problem (4 Marks)

Occurs due to repeated multiplication of derivatives:

$$\frac{\partial L}{\partial h_t} = \prod \tanh'(z)$$

Since derivatives < 1 , gradients vanish.

LSTM solution

Uses gates:

- forget gate
- input gate
- output gate

Cell state allows gradient flow across time.

2.2 Edge Device Recommendation (4 Marks)

Preferred model: **GRU**

Reasons:

- fewer parameters
- lower memory
- faster inference

2.3 Long Context NLP (4 Marks)

Preferred architecture: **LSTM**

Reasons:

- explicit memory cell
- better long-term dependency modeling

3 Question 2(B): LSTM Numerical (8 Marks)

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$

Given

$$C_{t-1} = 80$$

$$f_t = 0.85$$

$$i_t = 0.25$$

$$\tilde{C}_t = 40$$

Step 1:

$$0.85 \times 80 = 68$$

Step 2:

$$0.25 \times 40 = 10$$

Final:

$$C_t = 78$$

Interpretation

Forget gate dominates memory retention.

4 Question 3(A): Vanilla RNN (12 Marks)

$$h_t = \tanh(W_{hx}x_t + W_{hh}h_{t-1})$$

Parameters:

$$W_{hx} = 0.6$$

$$W_{hh} = 0.9$$

Inputs:

$$x_1 = 1, x_2 = 2, x_3 = -1$$

Initial state:

$$h_0 = 0$$

4.1 Step 1

$$h_1 = \tanh(0.6) = 0.537$$

4.2 Step 2

$$h_2 = \tanh(1.683) = 0.933$$

4.3 Step 3

$$h_3 = \tanh(0.240) = 0.235$$

5 Question 3(B): Optimizer Matching (8 Marks)

From the curve in the figure (page 4):

- Curve A SGD
- Curve B SGD + Momentum
- Curve C RMSProp
- Curve D Adam

6 Question 4(A): Overfitting Analysis (12 Marks)

Training accuracy = 99%

Validation accuracy = 72%

Problem: **Overfitting**

Evidence:

- large train-validation gap

- validation loss increasing

Dropout

Randomly disables neurons:

$$h_i = r_i a_i$$

Benefits:

- prevents co-adaptation
- improves generalization

7 Question 4(B): Weight Ratios (8 Marks)

Given symmetry:

$$w_1 = w_3, \quad w_2 = w_4$$

Gradients identical.

Thus ratios remain:

$$\frac{w_1}{w_3} = 1$$

$$\frac{w_2}{w_4} = 1$$

at all training steps and convergence.

8 Question 5: CNN Architecture (18 Marks)

Input:

$$128 \times 128 \times 3$$

Conv1 output:

$$O = \frac{128 - 5 + 4}{2} + 1 = 64$$

Output:

$$64 \times 64 \times 8$$

Conv2 parameters:

$$(3 \times 3 \times 8) \times 16 + 16 = 1168$$

Flatten:

$$2048$$

Fully connected:

$$2048 \times 4 + 4 = 8196$$

Stride vs pooling:

Stride reduces spatial size without extra operations.

9 Question 6: Transformer (15 Marks)

Embedding parameters:

$$|V| \times d_{model}$$

$$32000 \times 512$$

$$= 16,384,000$$

Feedforward layer:

$$512 \times 2048$$

Cross attention removal removes matrices:

$$W_Q, W_K, W_V$$

10 Question 7: Transformer Translation (11 Marks)

Self attention allows each word to attend to all other words.

Cross attention aligns source and target tokens.

Long sentence problem due to quadratic attention complexity.

Best solution: sliding window attention.

11 Complete TA Marking Rubric

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